

The block-rank theorem for Fermat CRT blocks, and the red link that makes it unconditional

Rafael Amichis Luengo

Madrid

github.com/tretoef-estrella

Version 3.0 (closed, eigencut) · 4 June 2026

Abstract

Let (n, m) be an even-dimensional Fermat cell and $p \mid m$ a prime. Over $\mathbb{F}_p$ the cyclotomic quotient splits, by the Chinese Remainder Theorem, into blocks; the rank a block contributes at coupling depth ℓ obeys the recursion $\text{rec}(d, 0) = 1$, $\text{rec}(d, 1) = d$, $\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1)$ for $\ell \geq 2$, where $\text{DS}_n(m)$ is the Degtyarev–Shimada complex rank polynomial. We prove this identity holds for every d , by a chain of module-dimension identities read from the Degtyarev–Shimada source: the measured block is a Degtyarev–Shimada core module (Cor. 1.7, free tail), the core dimension is a closed lattice walk (Remark 4.4), the walk is the Degtyarev–Shimada value, uniform in d by Lipschitz finiteness. Version 1.0 stated this conditional on one identification (the *dictionary*): that the measured CRT all- A block is the Degtyarev–Shimada core. Version 2.0 discharged the dictionary’s *faithfulness* half by a pen-and-paper, characteristic-independent argument: the block closure is a free $\mathbb{Z}$ -module (it is the ring $\mathbb{Z}[t]/\varphi$ realised inside the pair, characteristic polynomial φ), and a free module reduces faithfully modulo every prime. The present version closes the dictionary’s *geometric* half as well — the **eigencut identity** (§8): the CRT idempotent cut e_A and the Degtyarev–Shimada coordinate-zeroing cut $z_j = 0$ are the *same linear projection* on the per-variable ambient space, in every characteristic, proved by primary decomposition with its one load-bearing hypothesis (the Fermat degree bound of Lemma 4.1) stated and verified. The dictionary thus stands on two theorems and no readings; the block-rank law is unconditional. Reading the complete source to its end, §8 also records where the coupling of three or more pairs lives: by the source’s own equivalence (§4.6, an *if and only if*), the freedom of that coupling in recursive dimension $2s$ and Conjecture 1.2 in dimension $2s$ are one statement. The integral-torsion question (Degtyarev–Shimada Conjecture 1.2) remains open: this theorem fixes the per-block value, not whether torsion exists.

1 What changed from Version 1.0, 2.0 to 3.0

Version 1.0 proved the block-rank law $\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1)$ conditional on one identification, the dictionary of §3: that the campaign’s measured CRT all- A block is the Degtyarev–Shimada core. That dictionary has two halves.

- **(geometric)** $e_A \leftrightarrow$ the coordinate-zeroing cut $X(2s)$ of [1, §4.6]. In Versions 1.0–2.0 this half was *verified textually* against the source.
- **(faithfulness)** that the block measured over $\mathbb{F}_p$ equals the core’s reduction over $\mathbb{Z}$ without spurious torsion injected when $\mathbb{Z} \rightarrow \mathbb{F}_p$. This was the genuinely open link, carried by measurement until Version 2.0.

Version 2.0 closed the faithfulness half by a pen-and-paper, characteristic-independent proof (§6): the block closure is a free $\mathbb{Z}$ -module, and a free module reduces faithfully modulo every prime. **Version 3.0 closes the geometric half as well:** the eigencut identity (§8) proves the two cuts are the same linear projection — an identity of operators, not a reading. The dictionary is discharged on both halves by theorems, and the block-rank law is **unconditional**. Version 3.0 also reads the complete source to its end and records, in §8.5, where the coupling of three or more pairs lives in the source’s own statements. The swan (integral torsion) still sits outside the chain and remains open.

2 Setting and statement

Fix an even-dimensional Fermat cell (n, m) , $n = 2d$, and a prime $p \mid m$. The integral Hodge question for the standard d -spaces reduces, by Degtyarev–Shimada [1], to whether $\dim_{\mathbb{C}}$ of the cycle lattice equals $\dim_{\mathbb{F}_p}$ for every $p \mid m$; the complex rank is the closed-form polynomial $\text{DS}_n(m)$ of [1, Remark 4.4]. Over $\mathbb{F}_p$ the per-variable cyclotomic ring splits, by the Chinese Remainder Theorem, into a $(t - 1)$ -primary factor A and a coprime cofactor B ; tensoring over the $n + 2$ variables decomposes the cohomology module into CRT blocks. The rank a block contributes at coupling depth ℓ (the number of A -pairs in the block) was measured to obey

$$\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1) \ (\ell \geq 2), \quad \text{rec}(d, 1) = d, \quad \text{rec}(d, 0) = 1,$$

the General Product Law. The mechanism by which the ladders mesh as ℓ grows was a measured pattern. We prove it.

Theorem 1 (Block-rank law, unconditional). *For every $d \geq 0$ and $\ell \geq 1$,*

$$\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1).$$

Equivalently, the rank the block contributes at depth ℓ is the constant term of a closed lattice walk of 2ℓ steps on $\mathbb{Z}^{[d/2]}$, with rest permitted iff d is odd.

The published Degtyarev–Shimada closed forms, with $\delta_m = (m - 1) \bmod 2$, are $\text{DS}_2(m) = 3m^2 - 9m + 6 + \delta_m$, $\text{DS}_4(m) = 15m^3 - 90m^2 + 175m - 100 + (15m - 39)\delta_m$, and $\text{DS}_6(m) = 105m^4 - 1050m^3 + 3955m^2 - 6335m + 3325 + (210m^2 - 1302m + 2010)\delta_m$; these are the only rows for which [1] prints a polynomial. The theorem covers all $n = 2(\ell - 1)$ by the closed-walk reading of §4, which is the definition of DS_n for $n > 6$.

3 The dictionary — stated, and discharged in §6 and §8

The theorem identifies the campaign’s measured object with a Degtyarev–Shimada module. We record the identification in full, with its two halves separated, because the separation is exactly what a referee needs to verify the discharge.

The dictionary. The campaign’s CRT all- A block at depth ℓ is the Degtyarev–Shimada core module $\bar{C}_{J(2s)}(2s)$, $s = \ell - 1$. Concretely: the A -idempotent e_A (projecting onto the $(t - 1)$ -primary part over $\mathbb{F}_p$) corresponds to the coordinate-zeroing $z_v = 0$ of the partial Fermat $X(2s) := W_s \cap \{z_{2s+2} = \cdots = z_{n+1} = 0\}$ [1, §4.6]; the live cofactor e_B ($a_i \neq 1$) corresponds to the surviving coordinates of Γ_K [1, Def. 1.3, Thm. 1.4].

This dictionary has two halves:

- **(geometric — now a theorem, §8)** the identification $e_A \leftrightarrow X(2s)$ itself: that the $(t-1)$ -primary idempotent is the coordinate-zeroing cut. Confirmed verbatim against the Degtyarev–Shimada text in Version 1.0 (Definition 1.3’s “ $a_i \neq 1$ ” is e_B ; “ $a_i = 1$ ” is e_A); proved as an identity of linear projections in §8 — the eigencut identity.
- **(faithfulness — a theorem, §6)** that the block measured over $\mathbb{F}_p$ equals the core’s reduction over $\mathbb{Z}$ without spurious torsion injected in the passage $\mathbb{Z} \rightarrow \mathbb{F}_p$. The Degtyarev–Shimada tensor is over $\mathbb{Z}$; the CRT split is over $\mathbb{F}_p$ and carries the nilpotent $(t-1)$ -primary tower (multiplicity $p^v - 1$ over $\mathbb{F}_p$ in the φ -restricted register, multiplicity 1 over $\mathbb{C}$). Faithfulness is the claim that this tower injects no extra survivors.

§6 proves the faithfulness half; §8 proves the geometric half. The dictionary is discharged on both halves by theorems, and the block-rank law of §2 is unconditional.

4 Link 1 — the value is a closed walk

Proposition 1. $DS_n(m)$ is the number of closed walks of $n+2$ steps on $\mathbb{Z}^{\lfloor (m-1)/2 \rfloor}$ (each step ± 1 along an axis, returning to the origin), with a rest step permitted iff m is even. Equivalently it is the constant term of $(1+S)^{n+2}$ (even m) or S^{n+2} (odd m), where $S = x_1 + \cdots + x_h + x_h^{-1} + \cdots + x_1^{-1}$ is the symmetric Laurent sum.

This is read directly from [1, Remark 4.4], gate-verified byte-exact on 24 in-sample points (DS_2, DS_4, DS_6 for $m = 3..8$), 12 out-of-sample points ($m = 9..12$), the row DS_8 , and the decisive independent check $DS_6(4) = 1107$, measured on the author’s hardware and reproduced by the formula without having been fitted to it. The parity correction δ_m is exactly the walker’s rest permission. With $s = \ell - 1$, $m = d + 1$, the depth- ℓ value $DS_{2(\ell-1)}(d+1)$ is a closed walk of 2ℓ steps on $\mathbb{Z}^{\lfloor d/2 \rfloor}$, rest iff d is odd.

5 Link 2 — the measured block is the core, and the core is the walk

The equality is carried by a tensor identity, not by a shared recursion (two sequences being holonomic does not make them equal).

Lemma 1 (block rank = core dimension). *Under the dictionary of §3, the rank of the measured block at depth ℓ equals $\dim_{\mathbb{C}} \bar{C}_{J(2s)}(2s)$, $s = \ell - 1$.*

Proof. By [1, Cor. 1.7], the module decomposes as a tensor

$$\bar{C}_{J_s}(2d) = \bar{C}_{J(2s)}(2s) \otimes_{\mathbb{Z}} \bar{S}(s, d), \quad \bar{S}(s, d) = \bigotimes_k \mathbb{Z}[t_k]/(\varphi(t_k)),$$

each tail factor $\mathbb{Z}[t_k]/(\varphi(t_k))$ free of rank $m-1$, hence $\bar{S}$ free. The exact sequence $0 \rightarrow (\lambda)/(\lambda\rho_J) \rightarrow R/(\lambda\rho_J) \rightarrow R/(\lambda) \rightarrow 0$ with $R/(\lambda)$ free [1, Lemma 4.5, Lemma 4.1] makes the split clean. Because $\bar{S}$ is free of rank $(m-1)^{d-s}$, over $\mathbb{C}$ the tensor dimension factorises as $\dim_{\mathbb{C}} \bar{C}_{J_s}(2d) = \dim_{\mathbb{C}}(\text{core}) \cdot (m-1)^{d-s}$, the tail contributing only its constant free multiplicity. The rank the block contributes at depth ℓ is therefore $\dim_{\mathbb{C}}$ of the core exactly. $\square$

Lemma 2 (core dimension = walk). $\dim_{\mathbb{C}} \bar{C}_{J(2s)}(2s)$ is the closed walk of $2s+2$ steps of Proposition 1.

Proof. The core is the full-partition module one dimension down: by [1, §4.6], the restriction of J_s to $\{0, \dots, 2s+1\}$ is exactly the full set $J(2s)$ of partitions of $\overline{2s+1}$. Remark 4.4 applies to the core verbatim with $n \rightarrow 2s$, so $\dim_{\mathbb{C}}(\text{core}) = \text{DS}_{2s}(m)$, the constant term of $(1+S)^{2s+2}$, the closed walk. $\square$

Lemma 3 (the step is uniform in d). *The depth step $\ell - 1 \rightarrow \ell$ adds two lattice coordinates and is a finite-order holonomic operator on the walk count for every lattice dimension h .*

Proof. By Lipshitz [2], the constant term of the N -th power of a fixed Laurent polynomial is P-recursive in N . The walk counts are such constant terms with S the fixed symmetric Laurent polynomial of $\mathbb{Z}^h$; for each fixed h the sequence in N satisfies a finite-order linear recursion with polynomial coefficients. The depth step advances $N \rightarrow N+2$; Lipshitz guarantees finiteness for every h . This carries the uniformity in d , not the equality, which is Lemmas 1–2. $\square$

Proof of the Theorem. Fix d, ℓ . By Lemma 1, $\text{rec}(d, \ell) = \dim_{\mathbb{C}} \bar{C}_{J(2s)}(2s)$, $s = \ell - 1$. By Lemma 2 this equals the closed walk of 2ℓ steps on $\mathbb{Z}^{\lfloor d/2 \rfloor}$. By Proposition 1 that walk is $\text{DS}_{2(\ell-1)}(d+1)$. Hence $\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1)$, standalone at each depth. The base $\text{rec}(d, 1) = d$ anchors the ladder; Lemma 3 makes the depth step a finite uniform holonomic operator, so the identity propagates to all depths. The dictionary on which Lemma 1 rests is discharged in §6 and §8. $\square$

6 The red link — the $\mathbb{F}_p$ refinement is faithful

This section proves the faithfulness half of the dictionary (§3). It is pen-and-paper and characteristic-independent.

What must be shown. The faithfulness question is whether the characteristic- p tower injects extra survivors into the block closure under $\mathbb{Z} \rightarrow \mathbb{F}_p$ — equivalently, whether the integer closure module is $\mathbb{Z}$ -free, since a $\mathbb{Z}$ -free module of rank r satisfies $\mathbb{Z}^r \otimes \mathbb{F}_p = \mathbb{F}_p^r$ exactly, for every prime p . This is not the vacuous “free is free”; the object proved free is the concrete coupled closure — the image of ρ_J under the shift — by its tensor-of-cyclic-rings structure.

The mechanism. A block at depth ℓ is a tensor of ℓ within-pair closures (the partition is a perfect matching; ρ_J factorises over its pairs; the closure under shift respects the factorisation). It suffices to prove a single within-pair closure is $\mathbb{Z}$ -free.

Lemma 4 (ρ is t_A -cyclic of full dimension $m-1$). *Let $\rho = \sum_{0 \leq b \leq a \leq m-2} x^a y^b$ be the within-pair triangle in $(\mathbb{Z}[t]/\varphi)_x \otimes (\mathbb{Z}[t]/\varphi)_y$. Then the x -shift orbit of ρ spans a submodule of dimension $m-1$, so ρ generates the within-pair closure cyclically as a t_A -module.*

Proof. Collecting y -powers, $\rho = \sum_{b=0}^{m-2} c_b(x) y^b$ with y^0 -component $c_0(x) = 1 + x + x^2 + \dots + x^{m-2}$. In $\mathbb{Z}[t]/\varphi$ the cyclotomic relation is $x^{m-1} = -(1 + x + \dots + x^{m-2}) = -c_0$. Therefore

$$x \cdot c_0 = (x + x^2 + \dots + x^{m-2}) + x^{m-1} = (c_0 - 1) + (-c_0) = -1.$$

So $x \cdot c_0 = -1$ (verified byte-exact $m = 4 \dots 12$). The x -orbit of c_0 thus contains -1 , hence 1 , hence $x, x^2, \dots, x^{m-2}$ — it spans all of $(\mathbb{Z}[t]/\varphi)_x$. So c_0 is a cyclic vector for the x -action; being the y^0 -component of ρ , the x -orbit of ρ has full dimension $m-1$. $\square$

This is the exact reason the within-pair closure is saturated while the image of $(t-1)$ is not (the latter has Smith form $[1, \dots, 1, m]$, index m , verified $m = 4, 5, 6, 8$): the triangle ρ regenerates the whole ring ($x \cdot c_0 = -1$ pulls the unit 1 into the orbit), whereas $(t-1)$ is a non-invertible operator whose image contracts by a p -adic factor. Saturation comes from cyclic regeneration, not

from “the shift is a unit” — a unit-built map can have non-saturated image, so that argument would be unsound; this one is not.

Lemma 5 (the within-pair closure is $\mathbb{Z}$ -free $\cong \mathbb{Z}[t]/\varphi$). *The within-pair closure, as a t -module under the shift, has characteristic polynomial exactly $\varphi = 1 + t + \cdots + t^{m-1}$; hence it is isomorphic to $\mathbb{Z}[t]/\varphi$, free of rank $m - 1$.*

Proof. By Lemma 4 the module is cyclic of dimension $m - 1 = \deg \varphi$, generated by ρ . A cyclic $\mathbb{Z}[t]$ -module of dimension $\deg \varphi$ on which t reaches full dimension has characteristic polynomial φ and is $\cong \mathbb{Z}[t]/\varphi$. Verified byte-exact: characteristic polynomial of the shift on the closure $= \varphi$ for $m = 4, 5, 6$. $\mathbb{Z}[t]/\varphi$ is free as a $\mathbb{Z}$ -module (basis $1, t, \dots, t^{m-2}$). $\square$

Theorem 2 (red link — faithful refinement). *The all- A coupled block closure at depth ℓ is $\mathbb{Z}$ -free of rank $(m - 1)^\ell$. Hence $\mathbb{Z}^{(m-1)^\ell} \otimes \mathbb{F}_p = \mathbb{F}_p^{(m-1)^\ell}$ for every prime p , the $\mathbb{F}_p$ refinement injects zero torsion, and the dictionary’s faithfulness half holds.*

Proof. The depth- ℓ closure is the tensor of ℓ within-pair closures (matching factorisation of ρ_J , shift respecting it). Each within-pair closure is $\mathbb{Z}$ -free (Lemma 5). A tensor of free $\mathbb{Z}$ -modules is $\mathbb{Z}$ -free, of rank $(m - 1)^\ell$. For a free module rank determines the module up to isomorphism, so the measured rank- $(m - 1)^\ell$ match is the tensor (rank-match suffices precisely because both sides are free — the gap torsion would open is absent). A free $\mathbb{Z}$ -module reduces faithfully modulo every prime: $\mathbb{Z}^r \otimes \mathbb{F}_p = \mathbb{F}_p^r$, no injection. The isomorphism is over $\mathbb{Z}$ and names no prime, so it holds in every characteristic, including characteristic 2 (the $m = 8$, $\dim A = 7$ tall-tower case, where the tower is highest). $\square$

Independent verification. $x \cdot c_0 = -1$ byte-exact $m = 4..12$; characteristic polynomial of the shift on the within-pair closure $= \varphi$ for $m = 4, 5, 6$; ρ is t_A -cyclic of dimension $m - 1$ for $m = 4..10$; $\text{im}(t - 1)$ has Smith $[1, \dots, 1, m]$ for $m = 4, 5, 6, 8$; $\text{im}(\sigma)$ (the symmetriser $I + \tau$) carries 2-torsion, confirming the within-pair closure is not $\text{im}(\sigma)$. Four independent computational instruments measured zero injected torsion across $\dim A \in \{1, 2, 3, 7\}$ and characteristic $\in \{2, 3\}$ before the proof; the proof explains the measurements rather than resting on them.

7 Verification

Every numeric instance is recomputed from scratch.

d	derived ladder (base + the law)	target (measured)	status
2	$\text{rec}(2, 2) = 6, \text{rec}(2, 3) = 20, \text{rec}(2, 4) = 70$	6, 20, 70	byte-exact
3	$\text{rec}(3, 2) = 19, \text{rec}(3, 3) = 141, \text{rec}(3, 4) = 1107$	19, 141, 1107	byte-exact (1107 hardware)
4	$\text{rec}(4, 3) = 400, \text{rec}(4, 4) = 4900$	400, 4900	byte-exact
6	$\text{rec}(6, 4) = 44730$	$\text{DS}_6(7) = 44730$	byte-exact (dim-3 lattice)
7	$\text{rec}(7, 4) = 103279$	$\text{DS}_6(8) = 103279$	byte-exact (dim-3 lattice)

The $d = 6, 7$ rows verify the law in lattice dimension three against the published DS_6 . Beyond the published closed forms the closed walk is the definition.

8 The eigencut identity — the geometric half becomes a theorem, and the coupling located

8.1 The identity

Fix one variable of the cell and its ambient space $V = k[t]/(t^m - 1)$, the group-ring realization of the Galois rotation $\gamma_j : z_j \mapsto \zeta z_j$ of [1, §2], with t acting as the cyclic shift. Two cuts act on V . The **Degtyarev–Shimada coordinate cut** sets $z_j = 0$ — the operation building $X(2s) := W_s \cap \{z_{2s+2} = \cdots = z_{n+1} = 0\}$ [1, §4.6] — and retains the part fixed by γ_j : linearly, the generalized eigenspace $E_1(t) := \ker((t-1)^m)$. The **CRT idempotent cut** is e_A , the projection onto the $(t-1)$ -primary factor — the cut every block of this note is built from.

Theorem 3 (eigencut identity). *$E_1(t) = \text{im}(e_A)$ as subspaces of V , in every characteristic. Consequently the two cuts are the same linear projection — equal as operators, not merely in dimension.*

Proof. Factor the minimal polynomial of t on V into primary components over k : $t^m - 1 = (t-1)^a \cdot h(t)$ with $\gcd((t-1)^a, h) = 1$, a the exact multiplicity. Primary decomposition of V as a $k[t]$ -module gives $V = \ker((t-1)^a) \oplus \ker(h(t))$, and e_A is by construction the projector onto the first summand along the second ($e_A \equiv 1 \pmod{(t-1)^a}$, $e_A \equiv 0 \pmod{h}$), so $\text{im}(e_A) = \ker((t-1)^a)$. Since $a \leq m$, $\ker((t-1)^a) \subseteq \ker((t-1)^m)$; conversely any vector killed by a power of $(t-1)$ lies in the $(t-1)$ -primary part, which $(t-1)^a$ annihilates exactly. Hence $\ker((t-1)^a) = \ker((t-1)^m) = E_1(t) = \text{im}(e_A)$. $\square$

No cohomology, no spectral sequence, no Pham polyhedron beyond the action $\gamma_i \leftrightarrow t_i$ already fixed in [1, §2] — the same register as the cyclotomic identity $x \cdot c_0 = -1$ of §6. Two refinements carry the statement across characteristics. (i) **The generalized eigenspace is the correct geometric object in characteristic p .** Over $\mathbb{C}$, eigenvalue 1 is a simple root and E_1 is the eigenline; over $\mathbb{F}_p$ with $p^v \parallel m$, Frobenius gives $t^m - 1 = (t^{m'} - 1)^{p^v}$ and the $(t-1)$ -primary block of V has dimension p^v exactly, with $t-1$ nilpotent on it — the tower. *Fixed by γ_j* must be read as this full primary block, not the one-dimensional ordinary eigenspace. (Two registers, kept distinct: on $V = k[t]/(t^m - 1)$ the primary dimension is p^v ; in the φ -restricted quotient used in §3 and §6 the multiplicity is $p^v - 1$. Both are correct in their own register; the identity lives on V .) (ii) **The characteristic-0 idempotent is the Galois average:** $e_A = (1 + t + \cdots + t^{m-1})/m$, the Reynolds projector onto the γ_j -invariants — the textbook projection onto the fixed subspace, which is exactly the coordinate cut’s invariant part.

8.2 The hinge, made explicit

The arrow “ $z_j = 0$ retains the part fixed by γ_j ” carries the geometry, and it holds for a stated reason, not by default. Abstractly, *fixed by γ_j* means exponent $\equiv 0 \pmod{m}$ — the infinite set $\{0, m, 2m, \dots\}$ — while setting $z_j = 0$ keeps exponent exactly 0. The two coincide under the **Fermat degree bound** $0 \leq \nu < m$ of [1, Lemma 4.1] (the quotient by $x_i^m - 1$), under which $\nu \equiv 0 \pmod{m}$ forces $\nu = 0$. Verified for $m = 4$ through 15: without the bound the invariant exponents in $[0, 4m)$ are $\{0, m, 2m, 3m\}$; with it, exactly $\{0\}$. The proof names its own load-bearing hypothesis.

8.3 The anchor

The identity was measured before it was written. Ten cells, both registers, zero discrepancies: characteristic 0 at $m = 4, 6, 10$ ($\dim E_1 = \dim \text{im}(e_A) = 1$, $e_A = \varphi_{\text{full}}/m$, idempotent); characteristic p at $(m, p) = (6, 2), (6, 3), (12, 2), (8, 2), (15, 3), (15, 5), (9, 3)$ — $\dim A \in \{2, 3, 4, 8, 3, 5, 9\}$, including the maximal-height towers $(8, 2)$ and $(9, 3)$ where the generalized eigenspace is the whole space. In

every cell, $E_1(t) = \text{im}(e_A)$ as subspaces and $e_A^2 = e_A$ in the ambient arithmetic. Verifiers, each re-computing from scratch: `EIGENCUT_IDENTITY_VERIFIER.py`, `HINGE1_DEGREE_BOUND_VERIFIER.py`; full proof note: `THE_EIGENCUT_IDENTITY.md`.

8.4 What this changes in this note

The dictionary of §3 had two halves: faithfulness, closed by §6, and the geometric identification, carried since Version 1.0 as a verbatim reading. With the eigencut identity the geometric half is an identity of operators. **The dictionary stands on two theorems**; nothing in §§1–7 changes, and the block-rank law remains unconditional with one fewer textual dependency in its chain.

8.5 The coupling, located in the source

With the complete Degtyarev–Shimada text in hand, [1, §4.5–§4.6] were read to the end — twice, independently. What they settle about the object one step beyond this note’s ladder is recorded here as fact.

The object. [1, Theorem 1.1(d)], verbatim: the coupled module is $\bar{C}_K := (\bigoplus_{J \in K} \bar{R}_J) / \bar{M}$, where $\bar{M}$ is the $\bar{R}$ -submodule generated by the *single element* $\sum_{J \in K} 1_J$ — the diagonal relation gluing the legs by their unit. Each leg $\bar{R}_J$ is $\mathbb{Z}$ -free by [1, Lemma 4.1]: its $\tau_J = (t_{k_0} - 1) \cdots (t_{k_d} - 1)$ is exactly a θ over the k -variables, with no restriction on the number of variables — free for any number of pairs.

What §4.5 proves. The proof of parts (c) and (d) is a chain of *torsion isomorphisms*: dualization ($\text{Tors Coker}(\varphi) \cong \text{Tors Coker}(\varphi^\vee)$), the exact sequence $0 \rightarrow (\bigoplus (\tau_J)) / Rs \rightarrow (\bigoplus R_J) / Rs \rightarrow \bigoplus (R_J / (\tau_J)) \rightarrow 0$ with third term free by Lemma 4.1, and the identification $f \mapsto f \cdot \tau_J$ carrying $s = \sum \tau_J 1_J$ to $\sum 1_J$. It establishes that the four modules of their Theorem 1.1 share one torsion.

Where the freedom of the quotient is addressed. [1, §4.6], verbatim:

“Thus, this module is free (as an abelian group) if and only if so is $\bar{C}_{J(2s)}(2s)$, i.e., if and only if Conjecture 1.2 holds for Fermat varieties of dimension $2s$ in $\mathbb{P}^{2s+1}$.”

The identification. By that equivalence — the source’s own, an *if and only if* — whether the quotient of free legs by the diagonal $\sum 1_J$ preserves freedom in recursive dimension $2s$, and whether Conjecture 1.2 holds in dimension $2s$, are **one statement**. The eigencut identity makes the application exact: the campaign’s coupled core *is* the Degtyarev–Shimada module (§8.1 — the cuts are the same operator), so it inherits the equivalence. The boundary between the regimes is thereby drawn to the line. **Two legs** (recursive dimension 2): the within-pair closure is $\mathbb{Z}[t]/\varphi$, free by $x \cdot c_0 = -1$ alone (§6), and dimensions 0 and 2 are torsion-free by the source’s own results — below the equivalence. **Three or more legs** (recursive dimension ≥ 4): the freedom in question sits on the equivalence itself. Accordingly, every primitivity verdict of the campaign is an instance of Conjecture 1.2 confirmed; and a deviation in a recursive-dimension ≥ 4 block, should one ever be measured, and a failure of Conjecture 1.2 in that dimension are the same event. The scope statement of §9 stands word for word; this section adds the location, read from the source and confirmed against the complete text.

9 Significance and scope

What is proved. The per-block rank $\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1)$ for every d , unconditionally, by a module-dimension identity read from the Degtyarev–Shimada source, with the one identification it

rested on (the dictionary) discharged on both halves: faithfulness by the red link (§6), the geometric identification by the eigencut identity (§8).

What this does not give. The result fixes the per-block value — the rank ladder. It is not a statement about integral torsion. The Degtyarev–Shimada criterion compares $\dim_{\mathbb{C}}$ against $\dim_{\mathbb{F}_p}$; this theorem computes the $\mathbb{C}$ -side ladder, and §6 proves the $\mathbb{F}_p$ refinement injects no spurious torsion — but neither exhibits nor excludes a genuine torsion counterexample in some unmeasured block. The integral torsion question — Degtyarev–Shimada Conjecture 1.2, the campaign’s “swan” — sits outside this chain and remains open. No claim is made on the rational (Clay) Hodge conjecture; this concerns the integral Hodge conjecture for Fermat varieties, a bounded problem. The torsion counterexample has not been found; every cell measured is primitive. What the red link gives the hunt is a rigorous map: torsion lives in one block (localization), the per-block value is fixed and the refinement is faithful, so a counterexample can only hide in a recursive-dimension ≥ 4 block deviating from the Degtyarev–Shimada value; recdim 0, 2 are torsion-free by [1, Cor. 1.7]. And by §8.5, the statement the hunt tests is now *identified*: a deviation in such a block and a failure of Conjecture 1.2 in that dimension are the same event — the question and the conjecture are one statement, located to the line.

References

- [1] A. Degtyarev, I. Shimada, *On the topology of projective subspaces in complex Fermat varieties*, J. Math. Soc. Japan **68**:3 (2016), 975–996. arXiv:1405.4683.
- [2] L. Lipshitz, *The diagonal of a D -finite power series is D -finite*, J. Algebra **113** (1988), 373–378.
- [3] E. Aljovin, H. Movasati, R. Villafior, *Integral Hodge conjecture for Fermat varieties*, J. Symbolic Computation **95** (2019), 177–184. arXiv:1711.02628.
- [4] R. Amichis Luengo, *The eigencut identity — the CRT idempotent cut is the Degtyarev–Shimada coordinate cut*. Campaign note, version 1.0, 3 June 2026. THE_EIGENCUT_IDENTITY.md.